

Problem Statement

Load the load_digit dataset from sklearn.datasets.
This dataset is made up of 1797 8x8 images. Each image, is of a hand-written digit.
Therefore, each training example has 64 features (8X8) pixel values. Hence, the size of dataset is 1797X64.
Implement SVD in the following two ways:
(a) Step-by-step

(b) Using inbuilt SVD function in python

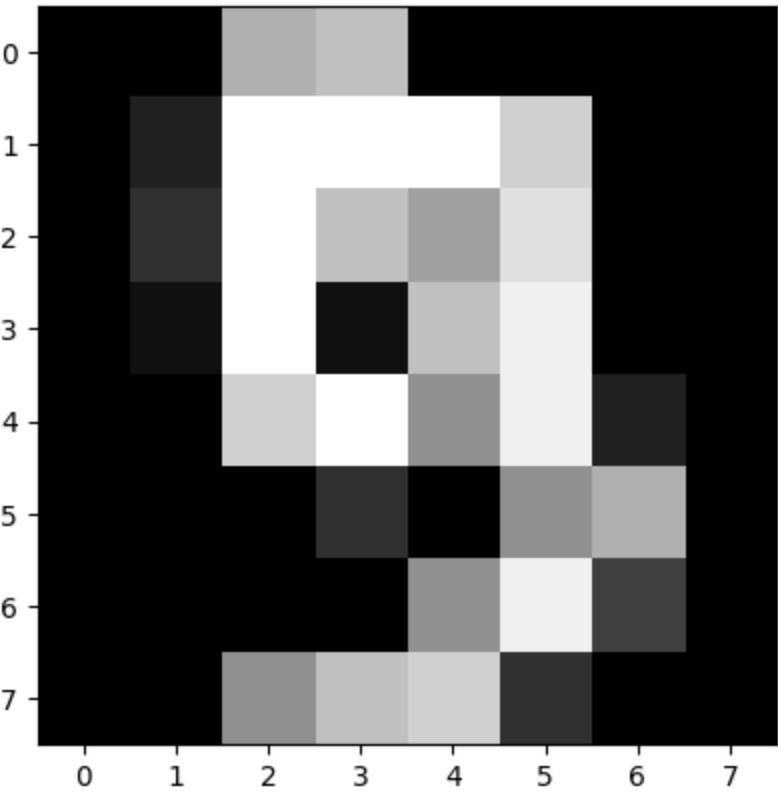
```
In [1]: from sklearn.datasets import load_digits
```

```
In [2]: digits = load_digits()
```

```
In [3]: import numpy as np
matrix = np.array(digits.data)
matrix,matrix.shape
```

Out[3]: (array([[0., 0., 5., ..., 0., 0., 0.],
[0., 0., 0., ..., 10., 0., 0.],
[0., 0., 0., ..., 16., 9., 0.],
...,
[0., 0., 1., ..., 6., 0., 0.],
[0., 0., 2., ..., 12., 0., 0.],
[0., 0., 10., ..., 12., 1., 0.]]),
(1797, 64))

```
In [4]: import matplotlib.pyplot as plt
plt.imshow(digits.images[9],cmap="grey")
plt.show()
```



```
In [5]: matrix_Transpose = np.transpose(matrix)
matrix_Transpose,matrix_Transpose.shape
```

```
Out[5]: (array([[ 0.,  0.,  0., ...,  0.,  0.,  0.],
               [ 0.,  0.,  0., ...,  0.,  0.,  0.],
               [ 5.,  0.,  0., ...,  1.,  2., 10.],
               ...,
               [ 0., 10., 16., ...,  6., 12., 12.],
               [ 0.,  0.,  9., ...,  0.,  0.,  1.],
               [ 0.,  0.,  0., ...,  0.,  0.,  0.]]),
        (64, 1797))
```

```
In [6]: aat = np.matmul(matrix,matrix_Transpose)
aat,aat.shape
```

```
Out[6]: (array([[3070., 1866., 2264., ..., 2812., 3006., 2898.],
               [1866., 4209., 3432., ..., 3906., 3083., 3307.],
               [2264., 3432., 4388., ..., 4005., 3063., 3697.],
               ...,
               [2812., 3906., 4005., ..., 5092., 3729., 4598.],
               [3006., 3083., 3063., ..., 3729., 4316., 3850.],
               [2898., 3307., 3697., ..., 4598., 3850., 4938.]]),
        (1797, 1797))
```

```
In [7]: values,vectors = np.linalg.eigh(aat)
vectors,vectors.shape
```

```
Out[7]: (array([[ -1.01350445e-06,  0.00000000e+00,  0.00000000e+00, ...,
                3.89296445e-02, -2.10250887e-03, -2.09114372e-02],
               [-6.34057303e-01,  1.25220381e-01, -3.66020270e-03, ...,
               -3.77989154e-02,  1.38656341e-02, -2.53199533e-02],
               [-1.02464218e-01,  5.14881721e-01,  2.27773937e-01, ...,
               -1.78272413e-02,  1.21951237e-02, -2.54559261e-02],
               ...,
               [-4.71205177e-02, -3.10177386e-02, -3.36495601e-02, ...,
               -1.16134751e-02,  1.87985701e-02, -2.98784403e-02],
               [ 2.80252919e-02, -5.50558356e-03, -2.71782002e-02, ...,
               2.34742370e-02, -8.66186959e-03, -2.67227437e-02],
               [-2.08054874e-02, -1.86070599e-03,  7.95443912e-03, ...,
               1.29922129e-02, -8.03379300e-04, -2.93865591e-02]]),
        (1797, 1797))
```

```
In [8]: values,len(values)
```

```
Out[8]: (array([-4.12203271e-10, -2.32084056e-10, -1.76093584e-10, ...,
                2.93769347e+05,  3.21485339e+05,  4.80977243e+06]),
        1797)
```

```
In [9]: values = np.clip(values,0,None)
values
```

```
Out[9]: array([      0.          ,      0.          ,      0.          , ...,
                293769.34713479,  321485.33927159,  4809772.4255891 ])
```

```
In [10]: desc = values[::-1]
desc
```

```
Out[10]: array([4809772.4255891 ,  321485.33927159,  293769.34713479, ...,
                0.          ,      0.          ,      0.          ])
```

```
In [11]: idx = np.argsort(values)
idx = idx[::-1]
idx
##we get indices for sorting
##np.argsort(values) returns a list for sorting in ascending
##we reverse the list to sort in desc
```

```
Out[11]: array([1796, 1795, 1794, ...,  611,  610,    0], dtype=int64)
```

```
In [12]: desc = values[idx]
desc
```

```
Out[12]: array([4809772.4255891 ,  321485.33927159,  293769.34713479, ...,
                0.          ,      0.          ,      0.          ])
```

```
In [13]: u = vectors[:,idx]
u
```

```
Out[13]: array([[ -2.09114372e-02, -2.10250887e-03,  3.89296445e-02, ...,
                0.00000000e+00,  0.00000000e+00, -1.01350445e-06],
               [-2.53199533e-02,  1.38656341e-02, -3.77989154e-02, ...,
               -1.83538207e-04, -8.97671633e-04, -6.34057303e-01],
               [-2.54559261e-02,  1.21951237e-02, -1.78272413e-02, ...,
               -5.82326445e-04, -3.29021569e-03, -1.02464218e-01],
               ...,
               [-2.98784403e-02,  1.87985701e-02, -1.16134751e-02, ...,
               -2.69964551e-02, -7.16040318e-03, -4.71205177e-02],
               [-2.67227437e-02, -8.66186959e-03,  2.34742370e-02, ...,
               -4.10253412e-02, -4.65719870e-03,  2.80252919e-02],
               [-2.93865591e-02, -8.03379300e-04,  1.29922129e-02, ...,
               3.69461696e-04, -1.11929470e-02, -2.08054874e-02]])
```

```
In [14]: sigma = np.zeros(matrix.shape)
```

```
In [15]: last_eigen_value_index = matrix.shape[1]
last_eigen_value_index
```

```
Out[15]: 64
```

```
In [16]: for i in range(0,last_eigen_value_index):
        sigma[i][i] = np.sqrt(desc[i])
sigma
```

```
Out[16]: array([[2193.11933683,  0.          ,  0.          , ...,  0.          ,
                0.          ,  0.          ],
               [ 0.          , 566.99677184,  0.          , ...,  0.          ,
                0.          ,  0.          ],
               [ 0.          ,  0.          , 542.00493276, ...,  0.          ,
                0.          ,  0.          ],
               ...,
               [ 0.          ,  0.          ,  0.          , ...,  0.          ,
                0.          ,  0.          ],
               [ 0.          ,  0.          ,  0.          , ...,  0.          ,
                0.          ,  0.          ],
               [ 0.          ,  0.          ,  0.          , ...,  0.          ,
                0.          ,  0.          ]])
```

```
In [17]: ata = np.matmul(matrix_Transpose,matrix)
ata,ata.shape
```

```
Out[17]: (array([[0.00000e+00, 0.00000e+00, 0.00000e+00, ..., 0.00000e+00,
                0.00000e+00, 0.00000e+00],
               [0.00000e+00, 1.64400e+03, 7.15400e+03, ..., 3.40600e+03,
                1.30600e+03, 6.60000e+01],
               [0.00000e+00, 7.15400e+03, 8.92850e+04, ..., 6.11890e+04,
                2.18770e+04, 4.72000e+03],
               ...,
               [0.00000e+00, 3.40600e+03, 6.11890e+04, ..., 1.44749e+05,
                5.32400e+04, 9.59500e+03],
               [0.00000e+00, 1.30600e+03, 2.18770e+04, ..., 5.32400e+04,
                3.77360e+04, 9.83300e+03],
               [0.00000e+00, 6.60000e+01, 4.72000e+03, ..., 9.59500e+03,
                9.83300e+03, 6.45300e+03]]),
        (64, 64))
```

```
In [18]: values,vectors = np.linalg.eigh(ata)
vectors,vectors.shape
```

```
Out[18]: (array([[ 1.00000000e+00,  0.00000000e+00,  0.00000000e+00, ...,
                  0.00000000e+00,  0.00000000e+00,  0.00000000e+00],
                [ 0.00000000e+00,  0.00000000e+00,  0.00000000e+00, ...,
                 -9.85740003e-03,  1.73619371e-02,  5.77192878e-03],
                [ 0.00000000e+00,  1.21707717e-15, -1.64953041e-14, ...,
                 -4.48071690e-02,  2.24200800e-01,  1.00696020e-01],
                ...,
                [ 0.00000000e+00, -1.14546632e-15, -2.30717158e-14, ...,
                  1.83210572e-01,  9.01286131e-02,  1.30885120e-01],
                [ 0.00000000e+00,  6.63464987e-16,  2.21243431e-16, ...,
                  2.14953332e-02,  3.68410903e-02,  3.92343661e-02],
                [ 0.00000000e+00, -3.57162986e-15, -1.39740166e-14, ...,
                 -6.40453868e-03,  1.15150706e-02,  6.76004954e-03]]),
         (64, 64))
```

```
In [19]: values, len(values)
```

```
Out[19]: (array([0.00000000e+00, 2.52923683e-12, 3.19114079e-11, 7.40483783e-01,
                  1.18769998e+00, 2.29473726e+00, 6.51802535e+00, 1.52320304e+01,
                  2.68560500e+01, 7.12408703e+01, 1.09105659e+02, 1.13894087e+02,
                  1.78092033e+02, 4.53277596e+02, 8.73520246e+02, 1.20868886e+03,
                  1.67263380e+03, 2.08149125e+03, 2.32172941e+03, 2.51877073e+03,
                  3.03690335e+03, 3.27211395e+03, 3.42595219e+03, 4.10076575e+03,
                  4.57731894e+03, 4.80069248e+03, 4.91796717e+03, 5.54416940e+03,
                  6.34448957e+03, 6.68199580e+03, 7.08239495e+03, 7.27368614e+03,
                  7.65227999e+03, 8.06923191e+03, 9.26122989e+03, 1.05839585e+04,
                  1.11895066e+04, 1.24304601e+04, 1.29144454e+04, 1.50373514e+04,
                  1.56141117e+04, 1.65923341e+04, 1.72537673e+04, 1.94152210e+04,
                  2.10061638e+04, 2.19837248e+04, 2.73737001e+04, 2.91890728e+04,
                  3.05385115e+04, 3.45170154e+04, 3.88137451e+04, 4.30961663e+04,
                  5.02498538e+04, 5.22834621e+04, 7.21026932e+04, 7.81520967e+04,
                  9.12489491e+04, 1.02640676e+05, 1.24763130e+05, 1.81129372e+05,
                  2.54168934e+05, 2.93769347e+05, 3.21485339e+05, 4.80977243e+06]),
         64)
```

```
In [20]: values = np.clip(values,0,None)
         values
```

```
Out[20]: array([0.00000000e+00, 2.52923683e-12, 3.19114079e-11, 7.40483783e-01,
                  1.18769998e+00, 2.29473726e+00, 6.51802535e+00, 1.52320304e+01,
                  2.68560500e+01, 7.12408703e+01, 1.09105659e+02, 1.13894087e+02,
                  1.78092033e+02, 4.53277596e+02, 8.73520246e+02, 1.20868886e+03,
                  1.67263380e+03, 2.08149125e+03, 2.32172941e+03, 2.51877073e+03,
                  3.03690335e+03, 3.27211395e+03, 3.42595219e+03, 4.10076575e+03,
                  4.57731894e+03, 4.80069248e+03, 4.91796717e+03, 5.54416940e+03,
                  6.34448957e+03, 6.68199580e+03, 7.08239495e+03, 7.27368614e+03,
                  7.65227999e+03, 8.06923191e+03, 9.26122989e+03, 1.05839585e+04,
                  1.11895066e+04, 1.24304601e+04, 1.29144454e+04, 1.50373514e+04,
                  1.56141117e+04, 1.65923341e+04, 1.72537673e+04, 1.94152210e+04,
                  2.10061638e+04, 2.19837248e+04, 2.73737001e+04, 2.91890728e+04,
                  3.05385115e+04, 3.45170154e+04, 3.88137451e+04, 4.30961663e+04,
                  5.02498538e+04, 5.22834621e+04, 7.21026932e+04, 7.81520967e+04,
                  9.12489491e+04, 1.02640676e+05, 1.24763130e+05, 1.81129372e+05,
                  2.54168934e+05, 2.93769347e+05, 3.21485339e+05, 4.80977243e+06])
```

```
In [21]: idx = np.argsort(values)
         idx = idx[::-1]
         idx
```

```
Out[21]: array([63, 62, 61, 60, 59, 58, 57, 56, 55, 54, 53, 52, 51, 50, 49, 48, 47,
                  46, 45, 44, 43, 42, 41, 40, 39, 38, 37, 36, 35, 34, 33, 32, 31, 30,
                  29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13,
                  12, 11, 10,  9,  8,  7,  6,  5,  4,  3,  2,  1,  0], dtype=int64)
```

```
In [22]: desc = values[idx]
         desc
```

```
Out[22]: array([4.80977243e+06, 3.21485339e+05, 2.93769347e+05, 2.54168934e+05,
1.81129372e+05, 1.24763130e+05, 1.02640676e+05, 9.12489491e+04,
7.81520967e+04, 7.21026932e+04, 5.22834621e+04, 5.02498538e+04,
4.30961663e+04, 3.88137451e+04, 3.45170154e+04, 3.05385115e+04,
2.91890728e+04, 2.73737001e+04, 2.19837248e+04, 2.10061638e+04,
1.94152210e+04, 1.72537673e+04, 1.65923341e+04, 1.56141117e+04,
1.50373514e+04, 1.29144454e+04, 1.24304601e+04, 1.11895066e+04,
1.05839585e+04, 9.26122989e+03, 8.06923191e+03, 7.65227999e+03,
7.27368614e+03, 7.08239495e+03, 6.68199580e+03, 6.34448957e+03,
5.54416940e+03, 4.91796717e+03, 4.80069248e+03, 4.57731894e+03,
4.10076575e+03, 3.42595219e+03, 3.27211395e+03, 3.03690335e+03,
2.51877073e+03, 2.32172941e+03, 2.08149125e+03, 1.67263380e+03,
1.20868886e+03, 8.73520246e+02, 4.53277596e+02, 1.78092033e+02,
1.13894087e+02, 1.09105659e+02, 7.12408703e+01, 2.68560500e+01,
1.52320304e+01, 6.51802535e+00, 2.29473726e+00, 1.18769998e+00,
7.40483783e-01, 3.19114079e-11, 2.52923683e-12, 0.00000000e+00])
```

```
In [23]: v = vectors[:,idx]
```

```
In [24]: v = np.transpose(v)
v
```

```
Out[24]: array([[ 0.00000000e+00,  5.77192878e-03,  1.00696020e-01, ...,
1.30885120e-01,  3.92343661e-02,  6.76004954e-03],
[ 0.00000000e+00,  1.73619371e-02,  2.24200800e-01, ...,
9.01286131e-02,  3.68410903e-02,  1.15150706e-02],
[ 0.00000000e+00, -9.85740003e-03, -4.48071690e-02, ...,
1.83210572e-01,  2.14953332e-02, -6.40453868e-03],
...,
[ 0.00000000e+00,  0.00000000e+00, -1.64953041e-14, ...,
-2.30717158e-14,  2.21243431e-16, -1.39740166e-14],
[ 0.00000000e+00,  0.00000000e+00,  1.21707717e-15, ...,
-1.14546632e-15,  6.63464987e-16, -3.57162986e-15],
[ 1.00000000e+00,  0.00000000e+00,  0.00000000e+00, ...,
0.00000000e+00,  0.00000000e+00,  0.00000000e+00]])
```

```
In [28]: sigma,desc
```

```
Out[28]: (array([[2193.11933683,  0.          ,  0.          , ...,  0.          ,
0.          ,  0.          ],
[ 0.          , 566.99677184,  0.          , ...,  0.          ,
0.          ,  0.          ],
[ 0.          ,  0.          , 542.00493276, ...,  0.          ,
0.          ,  0.          ],
...,
[ 0.          ,  0.          ,  0.          , ...,  0.          ,
0.          ,  0.          ],
[ 0.          ,  0.          ,  0.          , ...,  0.          ,
0.          ,  0.          ],
[ 0.          ,  0.          ,  0.          , ...,  0.          ,
0.          ,  0.          ]]),
array([4.80977243e+06, 3.21485339e+05, 2.93769347e+05, 2.54168934e+05,
1.81129372e+05, 1.24763130e+05, 1.02640676e+05, 9.12489491e+04,
7.81520967e+04, 7.21026932e+04, 5.22834621e+04, 5.02498538e+04,
4.30961663e+04, 3.88137451e+04, 3.45170154e+04, 3.05385115e+04,
2.91890728e+04, 2.73737001e+04, 2.19837248e+04, 2.10061638e+04,
1.94152210e+04, 1.72537673e+04, 1.65923341e+04, 1.56141117e+04,
1.50373514e+04, 1.29144454e+04, 1.24304601e+04, 1.11895066e+04,
1.05839585e+04, 9.26122989e+03, 8.06923191e+03, 7.65227999e+03,
7.27368614e+03, 7.08239495e+03, 6.68199580e+03, 6.34448957e+03,
5.54416940e+03, 4.91796717e+03, 4.80069248e+03, 4.57731894e+03,
4.10076575e+03, 3.42595219e+03, 3.27211395e+03, 3.03690335e+03,
2.51877073e+03, 2.32172941e+03, 2.08149125e+03, 1.67263380e+03,
1.20868886e+03, 8.73520246e+02, 4.53277596e+02, 1.78092033e+02,
1.13894087e+02, 1.09105659e+02, 7.12408703e+01, 2.68560500e+01,
1.52320304e+01, 6.51802535e+00, 2.29473726e+00, 1.18769998e+00,
7.40483783e-01, 3.19114079e-11, 2.52923683e-12, 0.00000000e+00]))
```

```
In [ ]:
```